Probability and Statistics

Discrete Random Variables and Probability Distributions

(PART 1)

曹圣彬
cccvolcano@m.scnu.edu.cn

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1. Random variable

2. Probability Distribution for Discrete Random Variable

3. Expected Value of Discrete Random Variable

4. Binomial Probability Distribution

1 Random variable

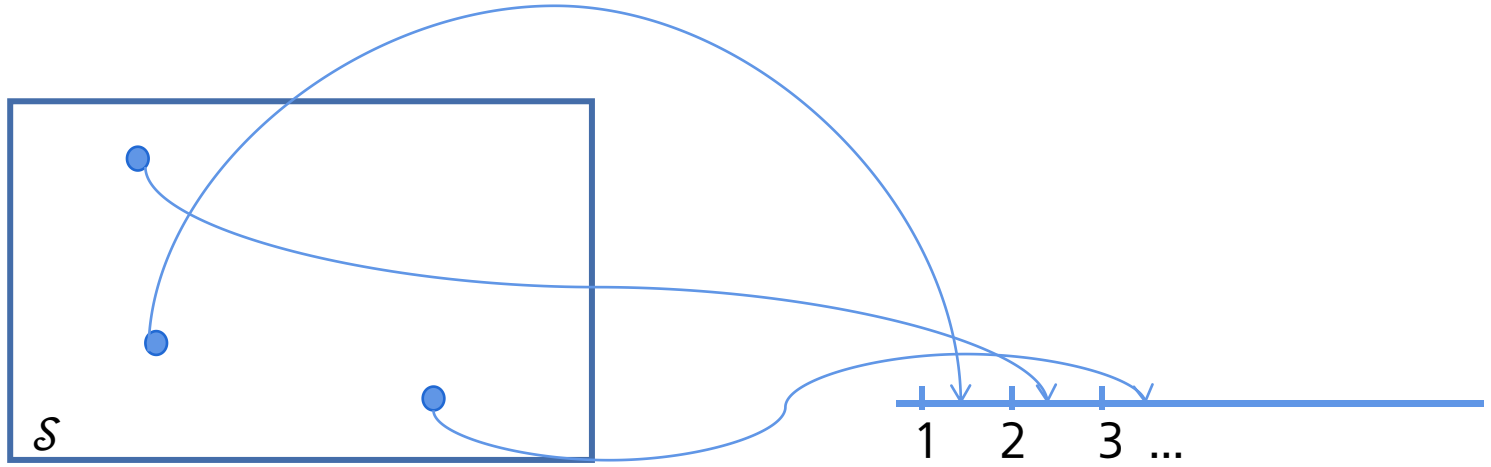
2 Probability Distribution for Discrete Random Variable

3 Expected Value of Discrete Random Variable

4 Binomial Probability Distribution

Random variable

- Random variable （随机变量）
  - Any rule that associates a number with each outcome in sample space
  - A function whose domain is the sample space and whose range is the set of real number
  - Denoted by uppercase letter, e.g., X, Y, Z
  - X(s) = x means that x is the value associated with the outcome s by the random variable X



Random variable

- Random variable （随机变量）
  - Two gas stations, each one has 6 gas pumps
  - X = the total number of pumps in use at two stations, Y = the difference between the number of pumps in use at two stations, Z = the maximum of number of pumps in use at two stations
  - s = (2, 3), X(s) = 5, Y(s) = 1, Z(s) = 3

		Second Station						
		0	**1**	**2**	**3**	**4**	**5**	**6**
First Station	**0**	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)	(0, 6)
	1	(1, 0)	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 0)	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 0)	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 0)	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 0)	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 0)	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

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  - s = (5, 4)



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		0	**1**	**2**	**3**	**4**	**5**	**6**
First Station	**0**	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)	(0, 6)
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  - s = (2, 3), X(s) = 5, Y(s) = 1, Z(s) = 3
  - s = (5, 4), X(s) = 9, Y(s) = 1, Z(s) = 5

		Second Station						
		0	**1**	**2**	**3**	**4**	**5**	**6**
First Station	**0**	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)	(0, 6)
	1	(1, 0)	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
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	3	(3, 0)	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
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Random variable

- Random variable （随机变量）
  - A worker continuously checks products coming from a product line until a defective one is found
  - Let E_i be the event that *i* number of products are checked
  - $E_1 = \{D\}$, $E_2 = \{N, D\}$, $E_3 = \{N, N, D\}$ ...

Random variable

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  - A worker continuously checks products coming from a product line until a defective one is found
  - Let E_i be the event that *i* number of products are checked
  - $E_1 = \{D\}$, $E_2 = \{N, D\}$, $E_3 = \{N, N, D\}$ ...
  - X = the number of productions that are checked
  - $X(E_1) = 1$, $X(E_2) = 2$, $X(E_3) = 3$ ...

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Random variable

- The relationship between random variable and event
  - A coin is tossed two times
  - $\mathcal{S} = \{HT, HH, TH, TT\}$
  - Let random variable X be the number of heads
  - What is the outcomes corresponding to X = 1?

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    - [X = 1] = {HT, TH}

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  - What is the outcomes corresponding to X = 1?
    - [X = 1] = {HT, TH}
  - What is the outcomes corresponding to X ≥ 1?

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Random variable

- The relationship between random variable and event
  - A coin is tossed two times
  - $\mathcal{S} = \{HT, HH, TH, TT\}$
  - Let random variable X be the number of heads
  - What is the outcomes corresponding to X = 1?
    - [X = 1] = {HT, TH}
  - What is the outcomes corresponding to X ≥ 1?
    - [X ≥ 1] = {HT, TH, HH}

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Random variable

- Bernoulli ramdom variable (伯努利随机变量)
  - Any random variable whose only possible values are 0 and 1
  - In coin tossing, $\mathcal{S} = \{H, T\}$
  - X = 1 if head side appears, X = 0 if tail side appears
  - X(H) = 1, X(T) = 0

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Random variable

- Types of random variable
  - Two types of variable for object in population
    - Discrete, continuous,
  - There are also two types for ramdom variable
    - Discrete random variable
      - Possible values either form a finite set or can be listed in an infinite sequence
    - Continuous random variable
      - Possible values consist of an entire interval of a line, or unions of disjoint lines, e.g., $[1, 2] \cup [4,5]$

1 Random variable

2 Probability Distribution for Discrete Random Variable

3 Expected Value of Discrete Random Variable

4 Binomial Probability Distribution

Probability Distribution for Discrete Random Variable

- Probability distribution （概率分布）
  - Probability for possible values of random variable
  - The probability distribution for discrete random variable is also called as probability mass function (PMF)
  - p(x) = the probability of the X value of x = P(X = x)
  - By the axiom of probability, $p(x) \geq 0$, $\sum_{x \in S} p(x) = 1$

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    - X = number of pumps in use in the first gas station

x	0	1	2	3	4	5	6
$p(x)$	.05	.10	.15	.25	.20	.15	.10

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    - X = number of pumps in use in the first gas station
    - P(X ≤ 2)

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    - X = number of pumps in use in the first gas station
    - $P(X \leq 2) = P(X = 0 \cup X = 1 \cup X = 2) = p(0) + p(1) + p(2) = 0.3$

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    - $P(X \geq 3)$

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    - X = number of pumps in use in the first gas station
    - $P(X \leq 2) = P(X = 0 \cup X = 1 \cup X = 2) = p(0) + p(1) + p(2) = 0.3$
    - $P(X \geq 3) = 1 - P(X \leq 2) = 0.7$

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$p(x)$	.05	.10	.15	.25	.20	.15	.10

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Probability Distribution for Discrete Random Variable

- There are 5 balls in a box, 2 red balls and 3 black balls, now you pick those balls one by one
- Y = number of picks until the first red ball
- What is PMF of Y?

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  - P(Y = 1) = P(one of two red balls is picked first) = 2 / 5 = 0.4

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  - P(Y = 2) = P(one of three black balls is picked first, then one of two red balls is picked second) = P(one of three black balls is picked first ∩ one of two red balls is picked second)

  = P(one of three black balls is picked first) * P(one of two red balls is picked second | one of three black balls is picked first) = 3 / 5 * 2 / 4 = 0.3

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  = P(one of three black balls is picked first) * P(one of two red balls is picked second | one of three black balls is picked first) = 3 / 5 * 2 / 4 = 0.3
  - P(Y = 3) = P(one of three black balls is picked first, then one of two black balls is picked second) * P(one of two red balls is picked third | one of three black balls is picked first, then one of two black balls is picked second) = 3 / 5 * 2 / 4 * 2 / 3

Probability Distribution for Discrete Random Variable

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- Y = number of picks until the first red ball
- What is PMF of Y?
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  = P(one of three black balls is picked first) * P(one of two red balls is picked second | one of three black balls is picked first) = 3 / 5 * 2 / 4 = 0.3
  - P(Y = 3) = 3 / 5 * 2 / 4 * 2 / 3 = 0.2
  - P(Y = 4) = 3 / 5 * 2 / 4 * 1 / 3 * 2 / 2 = 0.1

Probability Distribution for Discrete Random Variable

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- Y = number of picks until the first red ball
- What is PMF of Y?
  - P(Y = 1) = P(one of two red balls is picked first) = 2 / 5 = 0.4
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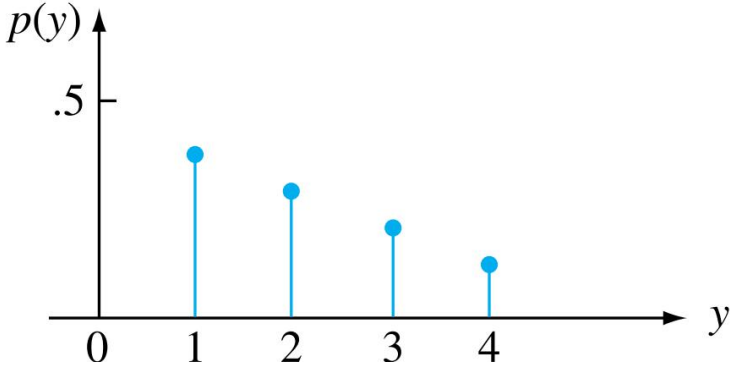
y	1	2	3	4
$p(y)$	.4	.3	.2	.1

Probability Distribution for Discrete Random Variable

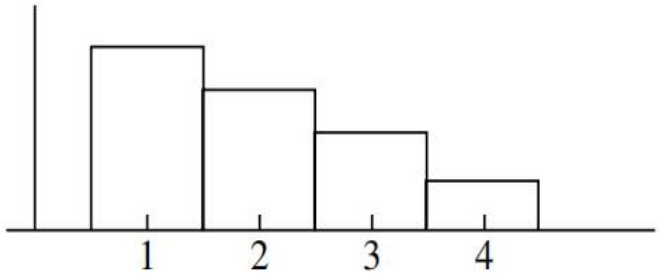
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  = P(one of three black balls is picked first) * P(one of two red balls is picked second | one of three black balls is picked first) = 3 / 5 * 2 / 4 = 0.3
  - P(Y = 3) = 3 / 5 * 2 / 4 * 2 / 3 = 0.2
  - P(Y = 4) = 3 / 5 * 2 / 4 * 1 / 3 * 2 / 2 = 0.1

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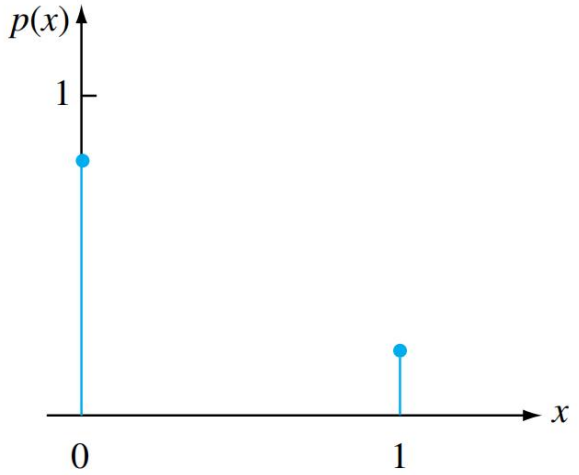
Line graph



Probability histogram

Probability Distribution for Discrete Random Variable

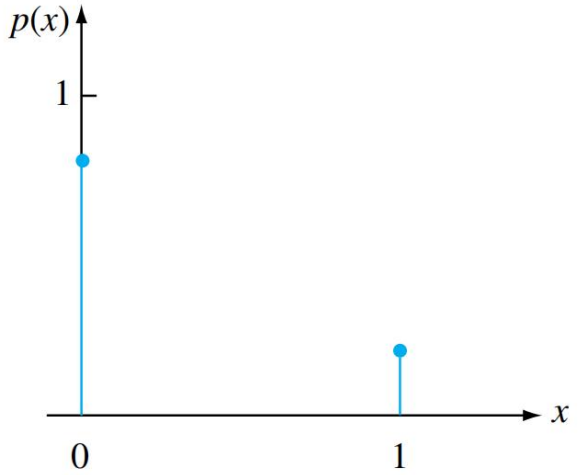
- Parameter of probability distribution
  - There are two types of computer, type A and B
  - X = 1 if a person buys type A, X = 0 otherwise
  - P(X = 1) = p(1) = α, p(0) = 1 - α
  - If α = 0.2, p(1) = 0.2, p(0) = 0.8



Probability Distribution for Discrete Random Variable

- Parameter of probability distribution
  - There are two types of computer, type A and B
  - X = 1 if a person buys type A, X = 0 otherwise
  - P(X = 1) = p(1) = α, p(0) = 1 - α
  - If α = 0.2, p(1) = 0.2, p(0) = 0.8
  - Different α provides different PMF of X
  - α is called parameter of the distribution
  - The collections of all probability distributions for differen

values of α is called a family of probability distributions



Probability Distribution for Discrete Random Variable

- Cumulative Distribution Function (CDF)（累积分布函数)
- The cumulative distribution function, denoted by F(x), of a random variable X is the probability of $X \leq x$
- $F(x) = P(X \leq x) = \sum_{i=1}^{x} P(X = i)$

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Probability Distribution for Discrete Random Variable

- Cumulative Distribution Function (CDF)（累积分布函数)
- There are 5 balls in a box, 2 red balls and 3 black balls, now you pick those balls one by one
- Y = number of picks until the first red ball
- F(1)

y	1	2	3	4
$p(y)$	.4	.3	.2	.1

Probability Distribution for Discrete Random Variable

- ## Cumulative Distribution Function (CDF)（累积分布函数)
- There are 5 balls in a box, 2 red balls and 3 black balls, now you pick those balls one by one
- Y = number of picks until the first red ball
- $F(1) = P(Y \leq 1) = P(Y = 1) = 0.4$

y	1	2	3	4
$p(y)$	.4	.3	.2	.1

Probability Distribution for Discrete Random Variable

- ## Cumulative Distribution Function (CDF)（累积分布函数)
- There are 5 balls in a box, 2 red balls and 3 black balls, now you pick those balls one by one
- Y = number of picks until the first red ball
- $F(1) = P(Y \leq 1) = P(Y = 1) = 0.4$
- F(2)

y	1	2	3	4
$p(y)$	.4	.3	.2	.1

Probability Distribution for Discrete Random Variable

- ## Cumulative Distribution Function (CDF)（累积分布函数)
- There are 5 balls in a box, 2 red balls and 3 black balls, now you pick those balls one by one
- Y = number of picks until the first red ball
- $F(1) = P(Y \leq 1) = P(Y = 1) = 0.4$
- $F(2) = P(Y \leq 2) = P(Y = 1) + P(Y = 2) = 0.4 + 0.3 = 0.7$

y	1	2	3	4
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Probability Distribution for Discrete Random Variable

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- $F(3)$

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- $F(3) = P(Y \leq 3) = P(Y = 1) + P(Y = 2) + P(Y = 3) = 0.4 + 0.3 + 0.2 = 0.9$

y	1	2	3	4
$p(y)$	.4	.3	.2	.1

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- $F(3) = P(Y \leq 3) = P(Y = 1) + P(Y = 2) + P(Y = 3) = 0.4 + 0.3 + 0.2 = 0.9$
- F(4)

y	1	2	3	4
$p(y)$	.4	.3	.2	.1

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- $F(3) = P(Y \leq 3) = P(Y = 1) + P(Y = 2) + P(Y = 3) = 0.4 + 0.3 + 0.2 = 0.9$
- $F(4) = P(Y \leq 4) = P(Y = 1) + P(Y = 2) + P(Y = 3) + P(Y = 4) = 0.4 + 0.3 + 0.2 + 0.1 = 1$

y	1	2	3	4
$p(y)$	.4	.3	.2	.1

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- $F(1) = P(Y \leq 1) = P(Y = 1) = 0.4$
- $F(2) = P(Y \leq 2) = P(Y = 1) + P(Y = 2) = 0.4 + 0.3 = 0.7$
- $F(3) = P(Y \leq 3) = P(Y = 1) + P(Y = 2) + P(Y = 3) = 0.4 + 0.3 + 0.2 = 0.9$
- $F(4) = P(Y \leq 4) = P(Y = 1) + P(Y = 2) + P(Y = 3) + P(Y = 4) = 0.4 + 0.3 + 0.2 + 0.1 = 1$
- F(2.7)



y	1	2	3	4
$p(y)$	.4	.3	.2	.1

Probability Distribution for Discrete Random Variable

- ## Cumulative Distribution Function (CDF)（累积分布函数)
- There are 5 balls in a box, 2 red balls and 3 black balls, now you pick those balls one by one
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- $F(2.7) = P(Y \leq 2.7) = F(2)$, the largest value less than 2.7

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Probability Distribution for Discrete Random Variable

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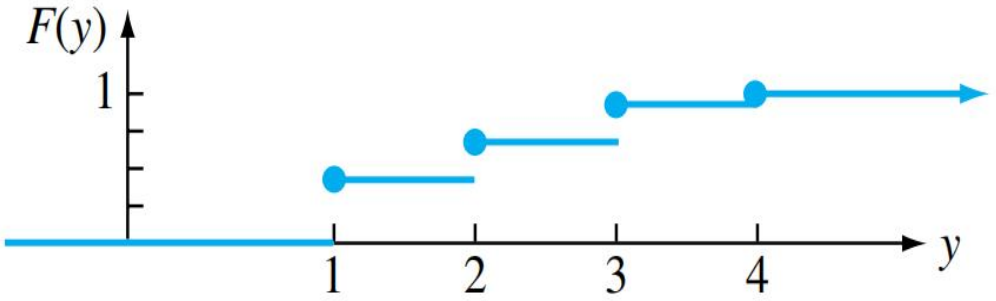
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Probability Distribution for Discrete Random Variable

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Probability Distribution for Discrete Random Variable

- ## Cumulative Distribution Function (CDF)（累积分布函数)
- A worker continuously checks products coming from a product line until a defective one is found, and assume P(D) = p
- X = the number of products that are checked
- PMF of X is

$$p(x) = \begin{cases} (1-p)^{x-1}p & x = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

- CDF of X is



Probability Distribution for Discrete Random Variable

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- CDF of X is

$$F(x) = \sum_{y \le x} p(y) = \sum_{y=1}^{x} (1-p)^{y-1}\, p = p \sum_{y=0}^{x-1} (1-p)^{y}$$

$$F(x) = p \cdot \frac{1-(1-p)^{x}}{1-(1-p)} = 1-(1-p)^{x}$$

$$F(x) = \begin{cases} 0 & x < 1 \\ 1-(1-p)^{[x]} & x \ge 1 \end{cases}$$

[x] is the largest integer ≤ x ([2.7] = 2)

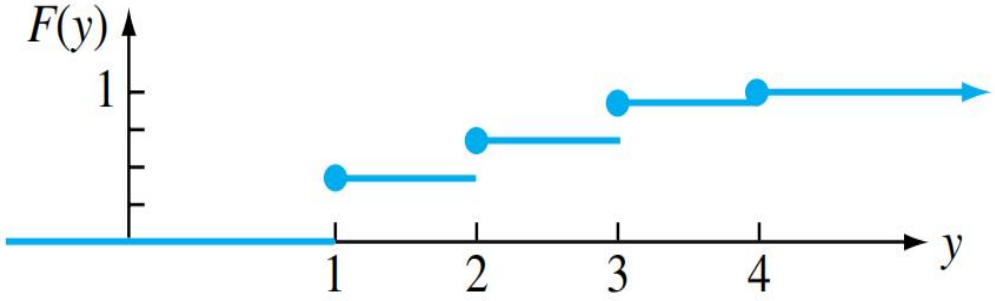
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Probability Distribution for Discrete Random Variable

- ## Cumulative Distribution Function (CDF)（累积分布函数)
- So far we derive CDF from PMF, yet we can derive PMF from CDF
- There are 5 balls in a box, 2 red balls and 3 black balls, now you pick those balls one by one
- Y = number of picks until the first red ball
- P(Y = 2)



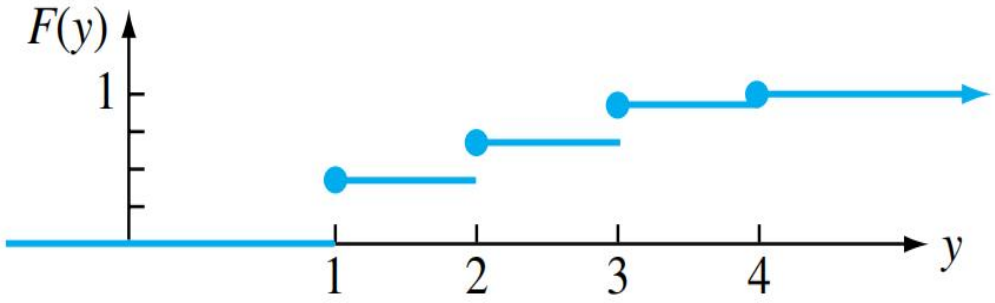
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Probability Distribution for Discrete Random Variable

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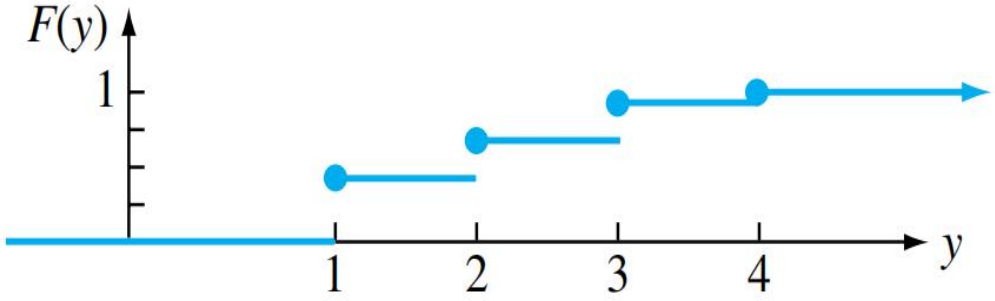


Probability Distribution for Discrete Random Variable

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- P(Y = 3)



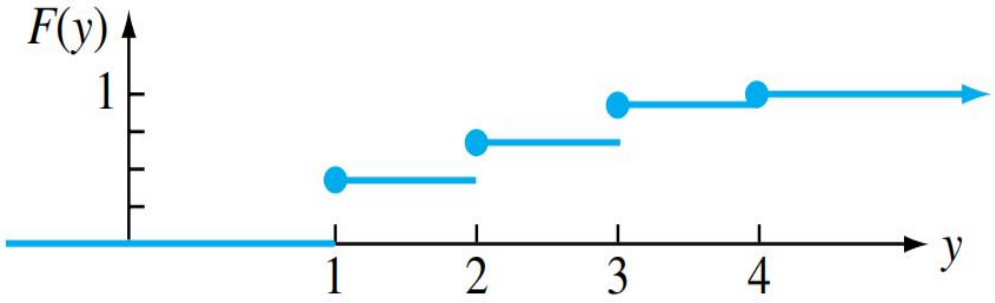
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Probability Distribution for Discrete Random Variable

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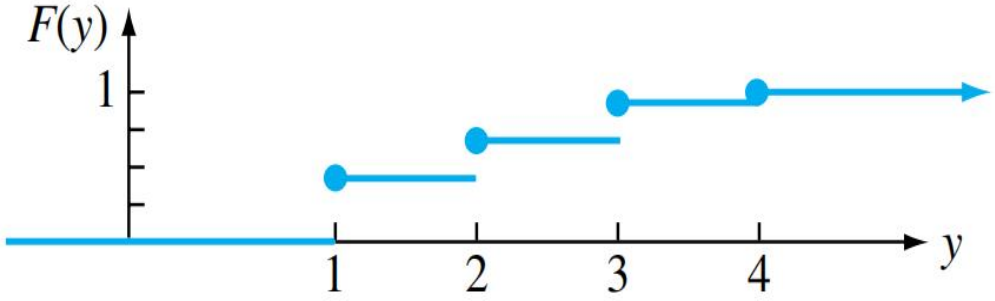
Probability Distribution for Discrete Random Variable

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- So far we derive CDF from PMF, yet we can derive PMF from CDF
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- $P(Y = 2) = F(2) - F(1) = P(Y = 1) + P(Y = 2) - P(Y = 1) = 0.7 - 0.4 = 0.3$
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- $P(2 \le Y \le 3)$



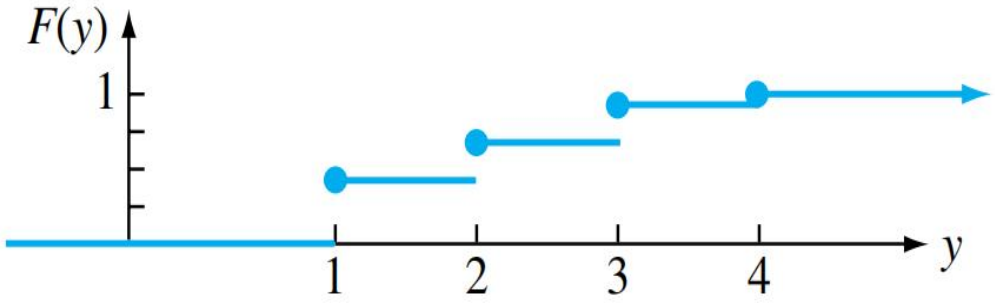
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Probability Distribution for Discrete Random Variable

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- $P(2 \le Y \le 3) = F(3) - F(1) = P(Y = 1) + P(Y = 2) + P(Y = 3) - P(Y = 1) = P(Y = 2) + P(Y = 3) = 0.5$

$$F(y) = \begin{cases} 0 & \text{if } y < 1 \\ .4 & \text{if } 1 \le y < 2 \\ .7 & \text{if } 2 \le y < 3 \\ .9 & \text{if } 3 \le y < 4 \\ 1 & \text{if } 4 \le y \end{cases}$$

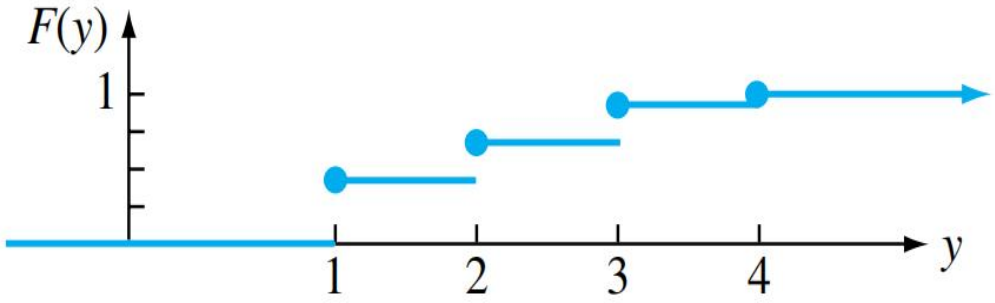


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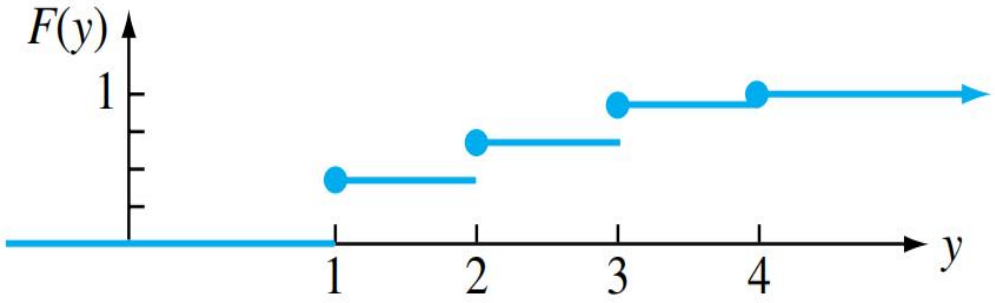


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Probability Distribution for Discrete Random Variable

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1 Random variable

2 Probability Distribution for Discrete Random Variable

3 Expected Value of Discrete Random Variable

4 Binomial Probability Distribution

Expected Value of Discrete Random Variable

- Expected value or mean (期望值)
- Let X be a discrete random variable with set of possible values D. The expected value or mean of X, denoted by $E(X)$, $E[X]$, μ_X, μ is
- $E(X) = \sum_{x \in D} x * p(x)$

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- $E(X) = \sum_{x \in D} x * p(x)$
- $E(X) = \frac{Sum\ of\ all\ outcomes}{Number\ of\ all\ outcomes} = \frac{\sum_{x=1}^{D} x*N(x)}{\sum_{i=1}^{D} N(x)} = \sum_{x=1}^{D} x * \frac{N(x)}{\sum_{i=1}^{D} N(x)} = \sum_{x=1}^{D} x * p(x)$
- $N(x)$ is the number of outcomes corresponding to $X = x$

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Expected Value of Discrete Random Variable

- Expected value (期望值)
- Consider a university having 15,000 students and let X = the number of courses for which a randomly selected student is registered
- E(X) =?

x	1	2	3	4	5	6	7
$p(x)$	.01	.03	.13	.25	.39	.17	.02

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Expected Value of Discrete Random Variable

- Expected value (期望值)
- Consider a university having 15,000 students and let X = the number of courses for which a randomly selected student is registered
- $E(X) = \sum_{i=1}^{7} i * p(i) = 4.57$

x	1	2	3	4	5	6	7
$p(x)$	.01	.03	.13	.25	.39	.17	.02

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Expected Value of Discrete Random Variable

- Massive machine type communications of 6G (mMTC)
- A successful transmission takes *s* seconds, a failure transmission takes *f* seconds, and idle slot takes *i* seconds
- Throughput of wireless network

- $\lim_{t\to\infty} \frac{\text{Number of successful transmissions}}{\text{t}}$

- If the number of nodes in the wireless network is *n*, the probability to transmit a signal in each slot is *p,* obtain the throughput
- $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \mathbb{I}(\text{i})}{\sum_{i=1}^{n} length\ of\ i\ th\ slot}$
- $\mathbb{I}(\text{i}) = 1$ if in *i* th slot there is one successful transmission
- $\mathbb{I}(\text{i}) = 0$ otherwise
- $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \mathbb{I}(\text{i}) \ / \text{n}}{\sum_{i=1}^{n} length\ of\ i\ th\ slot\ /\ n} = \frac{Probability\ of\ success}{Average\ length\ of\ slot}$
- Probability of success = P(s) = $C_{1,n} * p * (1-p)^{n-1} = n * p * (1-p)^{n-1}$
- Probability of idle slot = P(i) = $(1-p)^n$
- Probability of failure = 1 - P(s) - P(i)
- Average length of slot = s * P(s) + i * P(i) + f * P(f)
- $\frac{Probability\ of\ success}{Average\ length\ of\ slot} = \frac{n*p*(1-p)^{n-1}}{s*P(s)+i*P(i)+f*P(f)}$

Expected Value of Discrete Random Variable

- Expected value (期望值)
- A worker continuously checks products coming from a product line until a defective one is found, and assume P(D) = p
- N = the number of products that are checked
- PMF of N is

$$p_N(n) = p(1-p)^{n-1} \qquad n = 1, 2, \ldots$$

- What is E[N]?



Expected Value of Discrete Random Variable

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$$E[N] = \sum_n np_N(n) = p\sum_{n=1}^{\infty} n(1-p)^{n-1}$$

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$$\sum_{n=1}^{\infty}(1-p)^n = \frac{1}{1-(1-p)} - 1 = \frac{1}{p} - 1$$

$$\frac{\mathrm{d}}{\mathrm{d}p}\sum_{n=1}^{\infty}(1-p)^n = \sum_{n=1}^{\infty}\frac{\mathrm{d}}{\mathrm{d}p}(1-p)^n = -\sum_{n=1}^{\infty} n(1-p)^{n-1}$$

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$$E[N] = \sum_n n p_N(n) = p \sum_{n=1}^{\infty} n(1-p)^{n-1}$$

$$\sum_{n=1}^{\infty} (1-p)^n = \frac{1}{1-(1-p)} - 1 = \frac{1}{p} - 1$$

$$E[N] = -p \frac{\mathrm{d}}{\mathrm{d}p} \sum_{n=1}^{\infty} (1-p)^n = -p \frac{\mathrm{d}}{\mathrm{d}p} \left\{ \frac{1}{p} - 1 \right\} = -p \left\{ -\frac{1}{p^2} \right\} = \frac{1}{p}$$

$$\frac{\mathrm{d}}{\mathrm{d}p} \sum_{n=1}^{\infty} (1-p)^n = \sum_{n=1}^{\infty} \frac{\mathrm{d}}{\mathrm{d}p} (1-p)^n = -\sum_{n=1}^{\infty} n(1-p)^{n-1} = -p^{-2}$$

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Expected Value of Discrete Random Variable

- Expected value (期望值)
- Given a discrete random variable K whose PMF is given by

$$p_K(k) = \frac{5^k e^{-5}}{k!} \qquad k = 0, 1, 2, \ldots$$

- What is E[K]?

Expected Value of Discrete Random Variable

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$$p_K(k) = \frac{5^k e^{-5}}{k!} \qquad k = 0, 1, 2, \ldots$$

- What is E[K]?

$$E[K] = \sum_k k p_K(k) = e^{-5} \sum_{k=0}^{\infty} k \frac{5^k}{k!} = e^{-5} \sum_{k=1}^{\infty} \frac{5^k}{(k-1)!} = 5e^{-5} \sum_{k=1}^{\infty} \frac{5^{k-1}}{(k-1)!}$$

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Expected Value of Discrete Random Variable

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- What is E[K]?

$$E[K] = \sum_k k p_K(k) = e^{-5} \sum_{k=0}^{\infty} k \frac{5^k}{k!} = e^{-5} \sum_{k=1}^{\infty} \frac{5^k}{(k-1)!} = 5e^{-5} \sum_{k=1}^{\infty} \frac{5^{k-1}}{(k-1)!}$$

$$\sum_{i=0}^{\infty} \frac{a^i}{i!} = e^a \quad \text{Maclaurin series}$$

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Expected Value of Discrete Random Variable

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Expected Value of Discrete Random Variable

- Expected value of function of X
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Expected Value of Discrete Random Variable

- $Y = h(X) = 20 + 3 * X + 0.5 * X^2$
- The number of outcomes in the sample space is 10
- The number of outcomes corresponding to X = 4, N(X = 4) = 10 * 0.5 = 5
- The number of outcomes corresponding to X = 6, N(X = 6) = 10 * 0.3 = 3
- The number of outcomes corresponding to X = 8, N(X = 8) = 10 * 0.2 = 2

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- The number of outcomes corresponding to Y = 40, N(Y = 40) = N(X = 4) = 5
- Because Y = 40 comes from X = 4, which means there is one-to-one match, so the number of outcomes for Y = 40 equals to the number of outcomes for X = 4

x	4	6	8
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- The number of outcomes corresponding to Y = 76, N(Y = 76) = N(X = 8) = 2

- P(Y = 40) = 5 / (5 + 3 + 2) = P(X = 4) = 0.5
- P(Y = 56) = 3 / (5 + 3 + 2) = P(X = 6) = 0.3
- P(Y = 76) = 2 / (5 + 3 + 2) = P(X = 8) = 0.2

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- $E(Y) = \sum_y y * p(y) = \sum_x h(x) * p(x) = 40 * 0.5 + 56 * 0.3 + 76 * 0.2$

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Expected Value of Discrete Random Variable

- Expected value of linear function of X
- If Y = h(X) = a *X + b, what is E(Y)?



Expected Value of Discrete Random Variable

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Expected Value of Discrete Random Variable

- Expected value of linear function of X
- If Y = h(X) = a *X + b, what is E(Y)?
- E(Y) = a * E(X) + b
- $E(Y) = E(a*X + b) = \sum_x (a*x+b)*p(x) = \sum_x (a*x)*p(x) + \sum_x b*p(x)$
  $= a*\sum_x x*p(x) + b*\sum_x p(x) = a*E(x) + b$

Expected Value of Discrete Random Variable

- Expected value of linear function of X
- If Y = h(X) = a *X + b, what is E(Y)?
- E(Y) = a * E(X) + b
- E(Y) = E(a *X + b) = $\sum_x (a * x + b) * p(x) = \sum_x (a * x) * p(x) + \sum_x b * p(x)$
  $= a * \sum_x x * p(x) + b * \sum_x p(x) = a * E(x) + b$
- When b = 0, E(a * X) = $a * E(x)$
- When a = 0, E(b) = b

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Expected Value of Discrete Random Variable

- Variance (方差)
- Let X be a discrete random variable with set of possible values D, PMF $p(x)$, mean μ. The variance of X, denoted by $V(X)$, $V[X]$, σ_x^2, σ^2 is
- $V(X) = \sum_{x \in D} (x - \mu)^2 * p(x) = E[(x - \mu)^2]$

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Expected Value of Discrete Random Variable

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- $V(X) = \sum_{x \in D} (x - \mu)^2 * p(x) = E[(x - \mu)^2]$
- $V(X) = \frac{Sum\ of\ all\ squared\ deviations}{Number\ of\ all\ outcomes} = \frac{\sum_{x=1}^{D} (x-\mu)^2 * N(x)}{\sum_{i=1}^{D} N(x)} = \sum_{x=1}^{D} (x - \mu)^2 * \frac{N(x)}{\sum_{i=1}^{D} N(x)} = \sum_{x=1}^{D} (x - \mu)^2 * p(x)$
- $N(x)$ is the number of outcomes corresponding to $X = x$

Expected Value of Discrete Random Variable

- ## Variance (方差)
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- $N(x)$ is the number of outcomes corresponding to $X = x$
- ## Standard deviation（标准差）
- Denoted by σ_x, σ
- $\sigma_x = \sqrt{\sigma_x^2}$

Expected Value of Discrete Random Variable

- Computation of variance
- $V(X) = E[(X-\mu)^2] = E[X^2 - 2 * X * \mu + \mu^2] = E[X^2] - 2 * \mu * E[X] + \mu^2 = E[X^2] - 2 * \mu^2 + \mu^2 = E[X^2] - \mu^2$

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Expected Value of Discrete Random Variable

- Computation of variance
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- $E[X^n]$ is called n th moment of X
- $E[X]$ is called first moment of X
- $E[X^2]$ is called second moment of X

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Expected Value of Discrete Random Variable

- Variance (方差)
- A worker continuously checks products coming from a product line until a defective one is found, and assume P(D) = p
- N = the number of products that are checked
- PMF of N is

$$p_N(n) = p(1-p)^{n-1} \qquad n = 1, 2, \ldots$$

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$$\frac{d^2}{dp^2}\sum_{n=1}^{\infty}(1-p)^n = 2p^{-3}$$

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$$\sum_{n=1}^{\infty} n^2(1-p)^{n-1} = 2p^{-3}(1-p) + \sum_{n=1}^{\infty} n(1-p)^{n-1}$$

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- What is V[N]?

$$E[N^2] = \sum_{n=1}^{\infty} n^2\, p(1-p)^{n-1} = 2p^{-2}(1-p) + p^{-1} = \frac{2-p}{p^2}$$

$$\frac{d^2}{dp^2}\sum_{n=1}^{\infty}(1-p)^n = \sum_{n=1}^{\infty} n(n-1)(1-p)^{n-2} = \frac{1}{1-p}\sum_{n=1}^{\infty} n(n-1)(1-p)^{n-1} = 2p^{-3}$$

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$$\frac{d^2}{dp^2}\sum_{n=1}^{\infty}(1-p)^n = \sum_{n=1}^{\infty} n(n-1)(1-p)^{n-2} = \frac{1}{1-p}\sum_{n=1}^{\infty} n(n-1)(1-p)^{n-1} = 2p^{-3}$$

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Expected Value of Discrete Random Variable

- Variance of function of X
- If Y = h(X) = a *X + b, what is V(Y)?



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Expected Value of Discrete Random Variable

- Variance of function of X
- If Y = h(X) = a *X + b, what is V(Y)?
- $V(Y) = a^2 * \sigma_x^2$
- $V(Y) = E[(\ Y - E(Y)\ )^2] = E[(\ a\ * X\ +\ b - (a * \mu\ +\ b)\ )^2] = E[(a * (X - \mu))^2] = a^2 * [(X - \mu)^2]$

$= a^2 * \sigma_x^2$

1. Random variable

2. Probability Distribution for Discrete Random Variable

3. Expected Value of Discrete Random Variable

4. Binomial Probability Distribution

Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- An experiment consists of a sequence of *n* trials
- Each trial has two possible outcomes
  - For convenience, we denote the two oucomes as S and F
- The trials are independent
- The probability of S is fixed, denoted as p
- Binomial random variable $X \sim b(n, p)$ is the number of S's
- Its PMF is denoted by $b(x; n, p)$
- The order of those S's is not important

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- The number of trials in an experiments is *4*, the probability of success is *p*, and failure is *1 - p*

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Outcome	x	Probability	Outcome	x	Probability
SSSS	4	p^4	*FSSS*	3	$p^3(1-p)$
SSSF	3	$p^3(1-p)$	*FSSF*	2	$p^2(1-p)^2$
SSFS	3	$p^3(1-p)$	*FSFS*	2	$p^2(1-p)^2$
SSFF	2	$p^2(1-p)^2$	*FSFF*	1	$p(1-p)^3$
SFSS	3	$p^3(1-p)$	*FFSS*	2	$p^2(1-p)^2$
SFSF	2	$p^2(1-p)^2$	*FFSF*	1	$p(1-p)^3$
SFFS	2	$p^2(1-p)^2$	*FFFS*	1	$p(1-p)^3$
SFFF	1	$p(1-p)^3$	*FFFF*	0	$(1-p)^4$

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- When X = 3

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SSFF	2	$p^2(1-p)^2$	*FSFF*	1	$p(1-p)^3$
SFSS	3	$p^3(1-p)$	*FFSS*	2	$p^2(1-p)^2$
SFSF	2	$p^2(1-p)^2$	*FFSF*	1	$p(1-p)^3$
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Number of outcomes

Probability of each outcome

Outcome	*x*	Probability	Outcome	*x*	Probability
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SSSF	3	$p^3(1-p)$	*FSSF*	2	$p^2(1-p)^2$
SSFS	3	$p^3(1-p)$	*FSFS*	2	$p^2(1-p)^2$
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$$b(x; n, p) = C_{x,n} * p^x * (1-p)^{n-x}$$

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Binomial Probability Distribution

- Massive machine type communications of 6G (mMTC)
- A successful transmission takes *s* seconds, a failure transmission takes *f* seconds, and idle slot takes *i* seconds
- Throughput of wireless network

- $\lim_{t\to\infty} \frac{\text{Number of successful transmissions}}{\text{t}}$

- If the number of nodes in the wireless network is *n*, the probability to transmit a signal in each slot is *p,* obtain the throughput
- $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \mathbb{I}(\text{i})}{\sum_{i=1}^{n} length\ of\ i\ th\ slot}$
- $\mathbb{I}(\text{i}) = 1$ if in *i* th slot there is one successful transmission
- $\mathbb{I}(\text{i}) = 0$ otherwise
- $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \mathbb{I}(\text{i}) \ / \text{n}}{\sum_{i=1}^{n} length\ of\ i\ th\ slot\ /\ n} = \frac{Probability\ of\ success}{Average\ length\ of\ slot}$
- Probability of success = P(s) = $C_{1,n} * p * (1-p)^{n-1} = n * p * (1-p)^{n-1}$
- Probability of idle slot = P(i) = $(1-p)^{n}$
- Probability of failure = 1 - P(s) - P(i)
- Average length of slot = s * P(s) + i * P(i) + f * P(f)
- $\frac{Probability\ of\ success}{Average\ length\ of\ slot} = \frac{n*p*(1-p)^{n-1}}{s * P(s) + i * P(i) + f * P(f)}$

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- Four coins are tossed. If X = the number of heads, find PMF of X

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- $p_X(0)$

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- Four coins are tossed. If X = the number of heads, find PMF of X
- $p_X(0) = b(0;\ 4, 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$

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- $p_X(1)$

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- $p_X(0) = b(0;\ 4, 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$
- $p_X(1) = b(1;\ 4, 1/2) = C_{1,4} * (\frac{1}{2})^1 * (\frac{1}{2})^3 = 4/16$

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- $p_X(0) = b(0;\ 4, 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$
- $p_X(1) = b(1;\ 4, 1/2) = C_{1,4} * (\frac{1}{2})^1 * (\frac{1}{2})^3 = 4/16$
- $p_X(2)$

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- Four coins are tossed. If X = the number of heads, find PMF of X
- $p_X(0) = b(0;\ 4, 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$
- $p_X(1) = b(1;\ 4, 1/2) = C_{1,4} * (\frac{1}{2})^1 * (\frac{1}{2})^3 = 4/16$
- $p_X(2) = b(2;\ 4, 1/2) = C_{2,4} * (\frac{1}{2})^2 * (\frac{1}{2})^2 = 6/16$

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- Four coins are tossed. If X = the number of heads, find PMF of X
- $p_X(0) = b(0;\ 4, 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$
- $p_X(1) = b(1;\ 4, 1/2) = C_{1,4} * (\frac{1}{2})^1 * (\frac{1}{2})^3 = 4/16$
- $p_X(2) = b(2;\ 4, 1/2) = C_{2,4} * (\frac{1}{2})^2 * (\frac{1}{2})^2 = 6/16$
- $p_X(3)$

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- Four coins are tossed. If X = the number of heads, find PMF of X
- $p_X(0) = b(0;\ 4, 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$
- $p_X(1) = b(1;\ 4, 1/2) = C_{1,4} * (\frac{1}{2})^1 * (\frac{1}{2})^3 = 4/16$
- $p_X(2) = b(2;\ 4, 1/2) = C_{2,4} * (\frac{1}{2})^2 * (\frac{1}{2})^2 = 6/16$
- $p_X(3) = b(3;\ 4, 1/2) = C_{3,4} * (\frac{1}{2})^3 * (\frac{1}{2})^1 = 4/16$

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- Four coins are tossed. If X = the number of heads, find PMF of X
- $p_X(0) = b(0;\ 4, 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$
- $p_X(1) = b(1;\ 4, 1/2) = C_{1,4} * (\frac{1}{2})^1 * (\frac{1}{2})^3 = 4/16$
- $p_X(2) = b(2;\ 4, 1/2) = C_{2,4} * (\frac{1}{2})^2 * (\frac{1}{2})^2 = 6/16$
- $p_X(3) = b(3;\ 4, 1/2) = C_{3,4} * (\frac{1}{2})^3 * (\frac{1}{2})^1 = 4/16$
- $p_X(4)$

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Binomial Probability Distribution

- Binomial random variable (二项随机变量)
- Four coins are tossed. If X = the number of heads, find PMF of X
- $p_X(0) = b(0;\ 4,\ 1/2) = C_{0,4} * (\frac{1}{2})^0 * (\frac{1}{2})^4 = 1/16$
- $p_X(1) = b(1;\ 4,\ 1/2) = C_{1,4} * (\frac{1}{2})^1 * (\frac{1}{2})^3 = 4/16$
- $p_X(2) = b(2;\ 4,\ 1/2) = C_{2,4} * (\frac{1}{2})^2 * (\frac{1}{2})^2 = 6/16$
- $p_X(3) = b(3;\ 4,\ 1/2) = C_{3,4} * (\frac{1}{2})^3 * (\frac{1}{2})^1 = 4/16$
- $p_X(4) = b(4;\ 4,\ 1/2) = C_{4,4} * (\frac{1}{2})^4 * (\frac{1}{2})^0 = 1/16$

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Binomial Probability Distribution

- Mean and variance of binomial random variable $X \sim b(n, p)$

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Binomial Probability Distribution

- Mean and variance of binomial random variable $X \sim b(n, p)$
- $E[X] = n * p$, $Var[X] = n * p * (1 - p)$

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Binomial Probability Distribution

- Mean and variance of binomial random variable X ~ b(n, p)
- E[X] = n * p, Var[X] = n * p * (1 - p)

- $$E[X] = \sum_{x=0}^{n} x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^{n} x \frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}$$

$$= \sum_{x=1}^{n} \frac{xn(n-1)!}{(n-x)!x(x-1)!} p p^{x-1} (1-p)^{n-x}$$

$$= np \sum_{x=1}^{n} \frac{(n-1)!}{(n-x)!(x-1)!} p^{x-1} (1-p)^{n-x}$$

Let $j = x - 1$, which means that when $x = 1$, $j = 0$; and when $x = n$, $j = n - 1$. Thus, we obtain

$$np \sum_{j=0}^{n-1} \frac{(n-1)!}{(n-1-j)!j!} p^j (1-p)^{n-1-j} = np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} = np\{p + (1-p)\}^{n-1} = np(1)^{n-1}$$

$$= np$$

Binomial Probability Distribution

- Mean and variance of binomial random variable X ~ b(n, p)
- E[X] = n * p, Var[X] = n * p * (1 - p)

- E[X * (X - 1)]

$$= \sum_{x=0}^{n} x(x-1)\binom{n}{x}p^x(1-p)^{n-x}$$

$$= \sum_{x=0}^{n} \frac{x(x-1)n!}{(n-x)!x!}p^x(1-p)^{n-x}$$

$$= \sum_{x=2}^{n} \frac{x(x-1)n(n-1)(n-2)!}{(n-x)!x(x-1)(x-2)!}p^2p^{x-2}(1-p)^{n-x}$$

$$= n(n-1)p^2\sum_{x=2}^{n} \frac{(n-2)!}{(n-x)!(x-2)!}p^{x-2}(1-p)^{n-x}$$

Let $j = x-2$, which means that when $x = 2$, $j = 0$; and when $x = n$, $j = n-2$. Thus, we obtain

$$n(n-1)p^2\sum_{j=0}^{n-2} \frac{(n-2)!}{(n-2-j)!j!}p^j(1-p)^{n-2-j} = n(n-1)p^2\sum_{j=0}^{n-2}\binom{n-2}{j}p^j(1-p)^{n-2-j} = n(n-1)p^2\{p+(1-p)\}^{n-2}$$

$$= n(n-1)p^2$$

But E[X * (X-1)] = E[X^2] - E[X]

$$E[X^2] = n(n-1)p^2 + np$$

Var[X] = E[X^2] - ($E[X]$)^2

$$= np(1-p)$$

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Binomial Probability Distribution

- Mean and variance of binomial random variable $X \sim b(n, p)$
- One hundred balls are tossed into 50 boxes. What is the expected number of balls in the tenth box?

Binomial Probability Distribution

- Mean and variance of binomial random variable X ~ b(n, p)
- One hundred balls are tossed into 50 boxes. What is the expected number of balls in the tenth box?
- Let X denote the number of balls that go into the tenth box. Then X is a binomial random variable X = b(100, 1/50), hence E[X] = 100 * 1 / 50 = 2

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Binomial Probability Distribution

- Mean and variance of binomial random variable $X \sim b(n, p)$
- A coin is tossed 10 times. Given that there are 6 heads in total, what is the expected number of heads in the first 5 tosses?

Binomial Probability Distribution

- Mean and variance of binomial random variable $X \sim b(n, p)$
- A coin is tossed 10 times. Given that there are 6 heads in total, what is the expected number of heads in the first 5 tosses?
- The probability of a head is $6/10 = 0.6$. If Y denotes the number of heads in the first 5 tosses, then Y is a binomial random variable $Y = b(5, 0.6)$, hence $E[Y] = 5 \times 0.6 = 3$

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